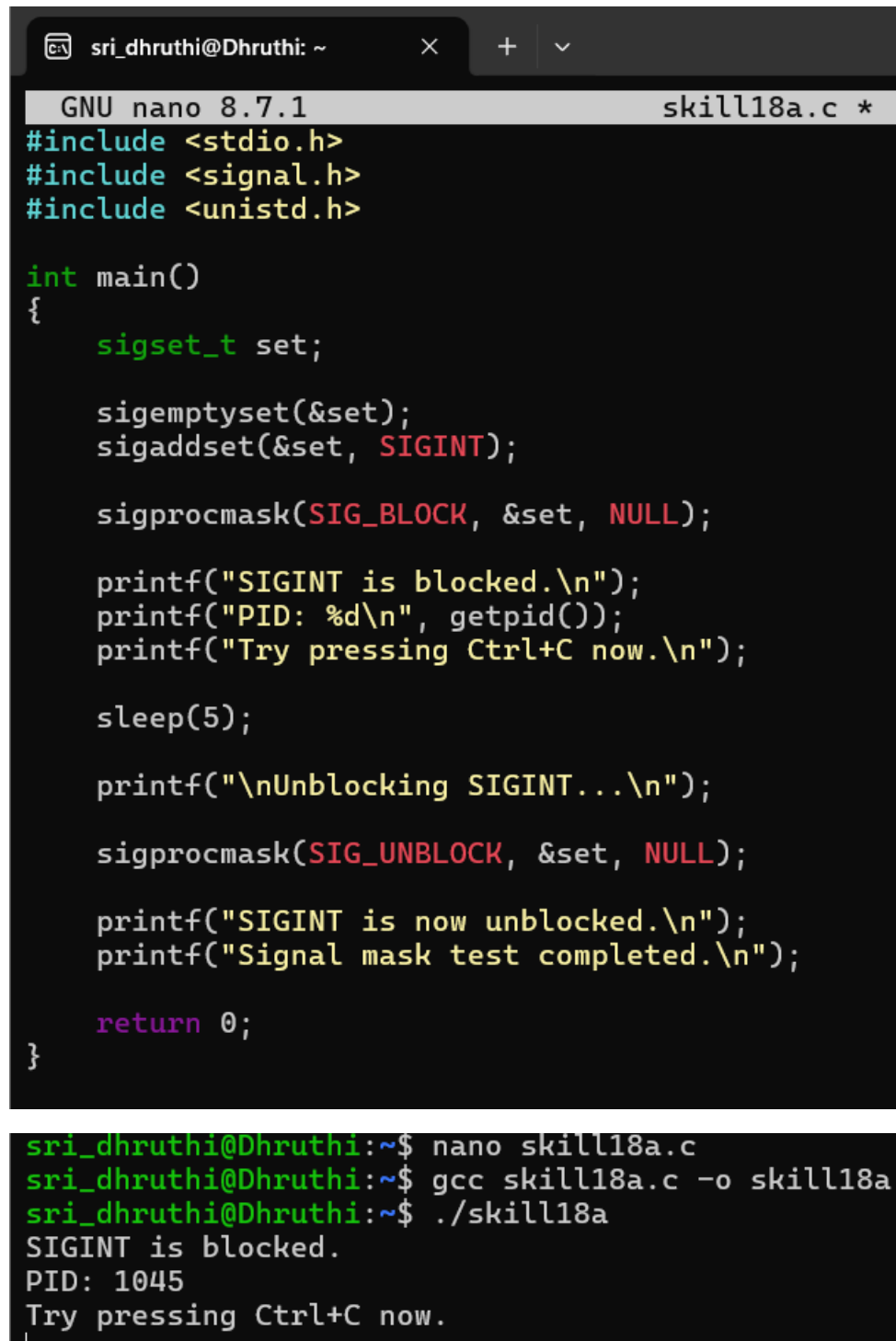


Os skill 18

18(a) Signal Delivery and Signal Masks

What it does: Demonstrates signal blocking using a signal mask and shows how signal behavior changes when the signal is unblocked.



```
sri_dhruthi@Dhruthi: ~  
GNU nano 8.7.1 skill18a.c *  
#include <stdio.h>  
#include <signal.h>  
#include <unistd.h>  
  
int main()  
{  
    sigset_t set;  
  
    sigemptyset(&set);  
    sigaddset(&set, SIGINT);  
  
    sigprocmask(SIG_BLOCK, &set, NULL);  
  
    printf("SIGINT is blocked.\n");  
    printf("PID: %d\n", getpid());  
    printf("Try pressing Ctrl+C now.\n");  
  
    sleep(5);  
  
    printf("\nUnblocking SIGINT...\n");  
  
    sigprocmask(SIG_UNBLOCK, &set, NULL);  
  
    printf("SIGINT is now unblocked.\n");  
    printf("Signal mask test completed.\n");  
  
    return 0;  
}  
  
sri_dhruthi@Dhruthi:~$ nano skill18a.c  
sri_dhruthi@Dhruthi:~$ gcc skill18a.c -o skill18a  
sri_dhruthi@Dhruthi:~$ ./skill18a  
SIGINT is blocked.  
PID: 1045  
Try pressing Ctrl+C now.
```

18(b) Resume Stopped Jobs

What it does: Stops a child process, sends SIGCONT to resume it, and verifies that the job completes successfully

```
sri_dhruthi@Dhruthi: ~  
GNU nano 8.7.1 skill18b.c *  
#include <stdio.h>  
#include <signal.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main()  
{  
    pid_t pid;  
    int status;  
  
    pid = fork();  
  
    if (pid < 0)  
    {  
        perror("fork");  
        return 1;  
    }  
  
    if (pid == 0)  
    {  
        printf("Job started. PID = %d\n", getpid());  
  
        raise(SIGSTOP);  
  
        printf("Job resumed.\n");  
        printf("Job completed successfully.\n");  
  
        return 0;  
    }  
  
    waitpid(pid, &status, WUNTRACED);  
  
    if (WIFSTOPPED(status))  
    {  
        printf("Job status: STOPPED\n");  
        printf("Sending SIGCONT to resume the job...\n");  
    }  
}
```

```
Signal mask test completed  
sri_dhruthi@Dhruthi:~$ nano skill18b.c  
sri_dhruthi@Dhruthi:~$ gcc skill18b.c -o skill18b  
sri_dhruthi@Dhruthi:~$ ./skill18b  
Job started. PID = 1084  
Job status: STOPPED  
Sending SIGCONT to resume the job...  
Job resumed.  
Job completed successfully.  
Job status: COMPLETED
```